KAMALESH J

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PROFESSIONAL SUMMARY

Third-year B.Tech Artificial Intelligence & Data Science student specializing in Computer Vision, Agentic AI, and full-stack AI systems. Experienced in building AI applications using Python, YOLO, OpenCV, MediaPipe, FastAPI, React, and LLM APIs, with hands-on work in video analytics, multi-agent systems, and backend-integrated applications.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS

AI / ML / GenAI: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Agentic AI, Multi-Agent Systems, Prompt Engineering, Data Preprocessing, Scikit-learn, TensorFlow, PyTorch

Computer Vision: YOLO, Ultralytics, OpenCV, MediaPipe, ByteTrack, InsightFace, Object Detection, Multi-Object Tracking, Face Detection, Face Mesh, Head Pose Estimation, Gaze Estimation, Video Analytics

Data Science & Analytics: Pandas, NumPy, Data Analysis, Data Cleaning, Data Visualization, K-Means Clustering, Power BI

Web & Backend: React, React Native, Flutter, FastAPI, Node.js, REST APIs, WebSockets, JWT Authentication, Streamlit

Databases: MongoDB, SQLite, Firebase

Tools & Core CS: Git, GitHub, VS Code, Google Colab, Kaggle, Roboflow, Vercel, Render, Object-Oriented Programming, Data Structures & Algorithms, DBMS

EXPERIENCE

Data Analytics Intern — Thiranex | 13 Jun 2026 - 12 Jul 2026

- **Built Python and SQL-based analytics workflows** using Pandas and NumPy to transform raw datasets into actionable business insights.
- **Developed Power BI dashboards** and performed exploratory analysis and visualization to identify trends and support business decision-making.

PROJECTS

EDITH — Multi-Agent AI Assistant

- Designed a 4-agent AI framework for intelligent routing, automation, self-healing, and plugin management.
- Configured **7** AI providers with **14** actively routed model configurations; validated successful use across **4** providers.
- Built 27 agent tools across 2 modules; validated with 49 automated test cases.
  Tech: Python, React, Agentic AI, Multi-Agent Systems, LLMs, Prompt Engineering

SentinelEye — AI Exam Proctoring System

- Developed an AI proctoring system using YOLO11, ByteTrack, MediaPipe, FastAPI, and MongoDB.
- Orchestrated multi-object tracking, event synchronization, risk evaluation, APIs, and WebSockets for up to 20 people simultaneously.
- Built a vision-to-backend pipeline for automated exam-event detection and risk assessment.
  Tech: Python, YOLO11, ByteTrack, MediaPipe, FastAPI, MongoDB

RESQ360 — Adaptive Emergency Response Platform

- Built a full-stack emergency response platform for SOS, public reporting, and real-time alerts.
- Enabled **150 m GPS verification**, 5-second alert polling, and **1 km / 30-minute** situation intelligence.
- Validated resource matching across **5 resource types** with **31/31 backend tests passed**.
  Tech: React Native, Expo, Node.js, GPS, OpenStreetMap, Leaflet, OSRM, Firebase

EDUCATION

B.Tech in Artificial Intelligence & Data Science | Sri Shakthi Institute of Engineering and Technology, Coimbatore | 2024 – 2028 | 3rd Year

CERTIFICATIONS & ACTIVITIES

- **AI-Machine Learning Engineer Certificate Course** — Reliance Foundation Skilling Academy / Skill India Digital | 150 Hours | 2026
- **HACKSPRINT'26** — 24-Hour Hackathon — Sri Shakthi Institute of Engineering and Technology | 2026
- **Hack Devengers 2.0 —** Participant **| Kaggle AI Agents Intensive Course —** Completed
- **11+ AI/ML & Full-Stack Projects** spanning Computer Vision, Agentic AI, Generative AI, and Data Analytics